

Fran Bartolić

Research Scientist | Google DeepMind

AI weather forecasting · New York, USA

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Experience

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|-----------------------|---|
| Jul 2026–
Present | Research Scientist Google DeepMind
New York, USA
Developing AI weather forecasting models for WeatherNext. |
| Dec 2022–
Jul 2026 | Research Scientist Salient Predictions
Remote (Croatia), then New York, USA <ul style="list-style-type: none">Developed probabilistic weather forecasting models across medium-range, subseasonal, and seasonal timescales.Developed methods for combining forecasts from multiple models, including neural-network approaches.Co-developed GEM-2 and GEM-3, efficient global weather models for probabilistic forecasting. |
| Jun–Dec 2021 | Research Resident Cervest
London, UK <ul style="list-style-type: none">Built probabilistic models of the urban heat island effect using geospatial data. |
| Feb–Jun 2020 | Research Analyst Center for Computational Astrophysics, Flatiron Institute
New York, USA <ul style="list-style-type: none">Developed a probabilistic model to reconstruct time-varying planetary surface maps from light curves using nonnegative matrix factorization and variational inference. |

Education

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| 2017–2023 | Ph.D. in Astrophysics University of St Andrews
St Andrews, Scotland <ul style="list-style-type: none">Developed Bayesian methods for gravitational microlensing and planetary surface mapping from astronomical time series.Created caustics, an open-source JAX package for differentiable computation of binary- and triple-lens microlensing light curves. |
| 2015–2017 | M.Sc. in Physics with Astrophysics University of Rijeka
Rijeka, Croatia <ul style="list-style-type: none">Conducted thesis research on circumbinary exoplanet dynamics at Lund Observatory, Sweden. |
| 2012–2015 | B.Sc. in Physics University of Split
Split, Croatia <ul style="list-style-type: none">Studied whether tidal engulfment explains the scarcity of close-in giant planets around evolved stars.Exchange semester at Lund University, Sweden. |

Publications

- 2026 S. Levang, **F. Bartolić**, T. Dickinson, C. Dwelle, P. Rauba, V. Cikojević. *Timestep-Conditioned Transformers for Global Weather Forecasting*, [arXiv:2608.06241](#).
- 2026 P. Rauba, V. Cikojević, **F. Bartolić**, S. Levang, T. Dickinson, C. Dwelle. *Probabilistic Transformers for Joint Modeling of Global Weather Dynamics and Decision-Centric Variables*, [arXiv:2601.03753](#).
- 2022 **F. Bartolić**, R. Luger, D. Foreman-Mackey, R. R. Howell, J. A. Rathbun. *Occultation mapping of Io's surface in the near-infrared I: Inferring static maps*, The Planetary Science Journal. doi:10.3847/PSJ/ac2a3e.
- 2021 R. Luger, E. Agol, **F. Bartolić**, D. Foreman-Mackey. *Analytic Light Curves in Reflected Light: Phase Curves, Occultations, and Non-Lambertian Scattering for Spherical Planets and Moons*, [arXiv:2103.06275](#).
- 2020 N. Golovich, W. Dawson, **F. Bartolić**, et al. *A Reanalysis of Public OGLE-III and IV Gravitational Microlensing Events*, [arXiv:2009.07927](#).
- 2018 V. Bozza, E. Bachelet, **F. Bartolić**, T. M. Heintz, A. R. Hoag, and M. Hundertmark. *VBBINARYLENSING: a public package for microlensing light-curve computation.*, 2018, MNRAS, 479, 5157. doi:10.1093/mnras/sty1791

Patents

- 2026 S. Levang, **F. Bartolić**. *Multi-model blending of probabilistic weather forecasts*, [US Patent Application 2026/0003100 A1](#) (filed 06/2025).
- 2026 S. Levang, **F. Bartolić**. *Multi-model blending via a neural network for probabilistic weather forecasts*, [US Patent Application 2026/0003101 A1](#) (filed 06/2025).
- 2026 S. Ridge, V. Cikojević, **F. Bartolić**. *Long-term weather forecasting framework using statistical and machine learning modeling*, [US Patent Application 2026/0104533 A1](#) (filed 10/2025).

Awards

- 2019 *Arthur Maitland Prize* for the best talk, University of St Andrews.
- 2015 *Dean's Award* for undergraduate academic excellence, University of Split.

Additional information

- Languages English (fluent), Croatian (native).
- Interests Cooking, reading, AI, history.